

How does the Internet **actually** work?

Summary notes by **Nishant Sir** — SIM se lekar DNS tak, poori journey ek data packet ki.



01 — GETTING ONLINE

SIM, ISP aur Broadband ke basics

Jab tum "internet chalate ho," asal mein tum ek real company se, real infrastructure use karne ka access khareed rahe ho.

`sim_card`

Chhota chip jo phone ko telecom company ke paas identify karta hai — ek ID card ki tarah.

`isp`

Jio, Airtel, Vi, BSNL — jo towers/cables ke maalik hain aur internet access bechte hain.

`mobile_tower`

Woh tall antenna structures — phone inse radio waves ke through baat karta hai.

`data`

Actual info jo travel karti hai — webpage, video, message. Measured in MB/GB.

`recharge`

ISP ko payment — fixed data + calling, fixed dino ke liye valid.

`broadband`

Ghar tak seedha wired (cable/fiber) connection — mobile tower ke bina.

`router_modem`

`wifi`

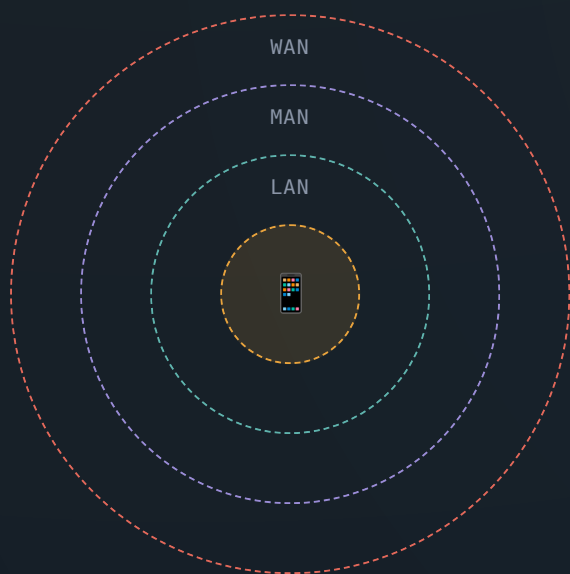
Broadband signal ko WiFi mein convert karne wala box, blinking lights ke saath.

Router se wireless connection — "connected but no internet" bhi possible hai!

02 — SCALE OF CONNECTIONS

Network ke 4 Types

Network matlab do ya zyada devices jo aapas mein baat kar sakte hain. Size ke hisaab se 4 categories.



Personal Area Network

PAN

Kuch meters ka range — phone se Bluetooth earbuds ya smartwatch.

Local Area Network

LAN

Ek ghar, classroom, ya office floor — jaise home WiFi ke sab devices.

Metropolitan Area Network

MAN

Poore city ya bade campus ki multiple LANs — jaise university ke sab departments.

Wide Area Network

WAN

Cities, states, countries ya poori duniya — ISP ka nationwide backbone.

03 — HOW REQUESTS TRAVEL

Client–Server Model

Har baar jab koi website khulti hai, ek "request → response" conversation ho rahi hoti hai.



Client

browser / app
– jo maangta

→ request: "give me homepage"

← response: HTML / video / data



Server

hai

data center —
jo deta hai

04 — FINDING A DEVICE

IP Address — Internet ka Phone Number

Jaise har insaan ka phone number unique hota hai, waise hi internet pe har device ka apna IP address hota hai.

domain	ip address
youtube.com	196.128.1.3
google.com	196.128.1.8
instagram.com	196.128.1.5
x.com	196.128.1.24
pw.com	196.128.1.4

Bilkul waise jaise phone mein "Varsha" naam save hota hai but asal mein call jaata hai uske number pe — waise hi "youtube.com" type karne pe request jaati hai uske IP address pe.

05 — THE PHONEBOOK LOOKUP

DNS (Domain Name System)

DNS ek giant, distributed phonebook hai jo domain name ko IP address mein convert karta hai.

01 you
Type karte ho **youtube.com** aur Enter dabate ho.

02 browser
DNS server se poochta hai: "youtube.com ka IP address kya hai?"

03 dns
Jawab deta hai: "142.250.183.78"

04 browser

Ab exactly pata hai ki Earth pe kis computer se baat karni hai.

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● ● ●

~ % nslookup google.com
Non-authoritative answer:
Name:    google.com
Address: 192.178.193.139

~ % nslookup youtube.com
Non-authoritative answer:
Name:    youtube.com
Address: 142.251.220.78
```

06 — ADDRESS FORMATS

IPv4 vs IPv6

Duniya mein devices itni badh gayi ki purana format khatam hone laga — isliye naya format aaya.

IPv4

142.250.183.78

- 4 numbers (0–255), dots se separated
- 32-bit — total ~4.3 billion unique addresses
- Purana, ab addresses khatam ho rahe hain

IPv6

2401:4900:1c20:abcd:0000:0000:0000:1a2b

- Hexadecimal + colons se banta hai
- 128-bit — undecillion addresses possible
- Naya format, shortage solve karta hai

Quick Summary

Term	One-liner
SIM Card	Phone ko telecom company ke paas identify karta hai
ISP	Company jo internet infrastructure aur access bechti hai
PAN / LAN / MAN / WAN	Personal → Local → City → Global scale ka network

Client / Server	Client maangta hai, Server jawab deta hai
IP Address	Har device ka unique internet "phone number"
DNS	Domain name ko IP address mein convert karta hai
IPv4 vs IPv6	32-bit purana format vs 128-bit naya format